Yu-Han Wu □□□

PhD student in diffusion models and machine learning theory

LPSM, Sorbonne University · Google DeepMind · Paris, France

yhwu@google.com
yh-wu.com
github.com/pojoowu
Google Scholar

EDUCATION

PhD in applied mathematics and machine learning — 2024 – 2027 (expected)
Sorbonne University (LPSM) and Google DeepMind, Paris
Advisors: Gérard Biau, Claire Boyer and Pierre Marion; Quentin Berthet and Romuald Elie at Google DeepMind.

Diploma of the École Normale Supérieure, applied mathematics — 2024
École Normale Supérieure, Paris

MSc in statistics and machine learning — 2023
Université Paris-Saclay

BSc in applied mathematics — 2021
Université Paris Cité

Classe préparatoire (MPSI/MP) — 2020
Lycée Louis-le-Grand, Paris

PUBLICATIONS

CONFERENCE AND JOURNAL PAPERS

1. **Yu-Han Wu**, Quentin Berthet, Gérard Biau, Claire Boyer, Romuald Elie and Pierre Marion (2026). Optimal Stopping in Latent Diffusion Models. In International Conference on Machine Learning (ICML).
   Oral at the PriGM workshop, EurIPS 2025
2. **Yu-Han Wu**, Pierre Marion, Gérard Biau and Claire Boyer (2025). Taking a Big Step: Large Learning Rates in Denoising Score Matching Prevent Memorization. In Proceedings of the Thirty Eighth Conference on Learning Theory (COLT), PMLR 291:5718–5756.
3. Pierre Marion, **Yu-Han Wu**, Michael E. Sander and Gérard Biau (2024). Implicit regularization of deep residual networks towards neural ODEs. In The Twelfth International Conference on Learning Representations (ICLR).
   Spotlight
4. David Conlon and **Yu-Han Wu** (2023). More on lines in Euclidean Ramsey theory. Comptes Rendus. Mathématique, 361(G5), 897–901.

PREPRINTS

1. Guillaume Couairon, Alexis Jacq, **Yu-Han Wu**, Renu Singh, Yana Hasson, Quentin Berthet and Romuald Elie (2026). Kastor: An efficient fine-tuning strategy for generative emulation of PDE simulations. arXiv:2608.06107.
2. DiffusionGemma Team (2026). DiffusionGemma Technical Report. arXiv:2608.00146.
3. Pierre Marion and **Yu-Han Wu** (2026). Understanding diffusion models requires rethinking (again) generalization. arXiv:2605.06077.
4. Quentin Berthet, **Yu-Han Wu**, Clément Crepy, Romuald Elie, Klaus Greff and Michael E. Sander (2026). MIND: Monge Inception Distance for Generative Models Evaluation. arXiv:2605.06797.

TALKS AND POSTERS

Invited talk at Optimisation et Apprentissage appliqués aux contenus numériques — Jun 2026
MathSTIC, USPN, France

Poster at Lemanth 2026 Apr 2026
EPFL Bernoulli Center, Lausanne, Switzerland

Oral at PriGM Workshop, EurIPS 2025 Dec 2025
Copenhagen Bella Center, Copenhagen, Denmark

Talk at COLT 2025 Jul 2025
ENS Lyon, Lyon, France

Talk at Journées de Statistique 2025 Jul 2025
Université Aix-Marseille, Marseille, France

DISTINCTIONS

Oral at the PriGM workshop, EurIPS 2025 2025
Optimal Stopping in Latent Diffusion Model

Spotlight, ICLR 2024 2023
Implicit regularization of deep residual networks towards neural ODEs

RESEARCH EXPERIENCE

Research intern Winter 2023
Owkin, Paris
Federated learning and invariant risk minimization. Supervisor Ulysse Marteau-Ferey.

Research intern Summer 2023
Sorbonne University (LPSM)
Implicit regularization of residual networks. Supervisors Gérard Biau and Pierre Marion.

Research intern Summer 2022
California Institute of Technology
Extremal combinatorics and Euclidean Ramsey theory. Supervisor David Conlon.

SERVICE

Reviewer, ICML 2026

Reviewer, NeurIPS 2026

Reviewer, PriGM workshop at EurIPS 2025 – 2026

SKILLS

Mathematics probability, statistics, optimization, stochastic analysis

Programming Python (JAX, PyTorch, NumPy), LaTeX, Git